OneAgent and ActiveGate Migration

> **Series:** M2S | **Notebook:** 6 of 8 | **Created:** January 2026 | **Last Updated:** 01/30/2026

Agent migration is the core technical step. OneAgents must be reconfigured to communicate with SaaS instead of Managed.

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Prerequisites

Before starting this notebook, you should have:

Requirement	Description
Completed M2S-01 to M2S-05	Configurations migrated
SaaS tenant ready	Tenant provisioned and accessible
Network connectivity	Hosts can reach SaaS or ActiveGate
Deployment tokens	SaaS installer tokens created
Maintenance window	Scheduled for migration

Learning Objectives

By the end of this notebook, you will:

- Deploy Environment ActiveGates for SaaS
- Understand OneAgent migration strategies
- Execute OneAgent reconfiguration
- Validate successful agent migration

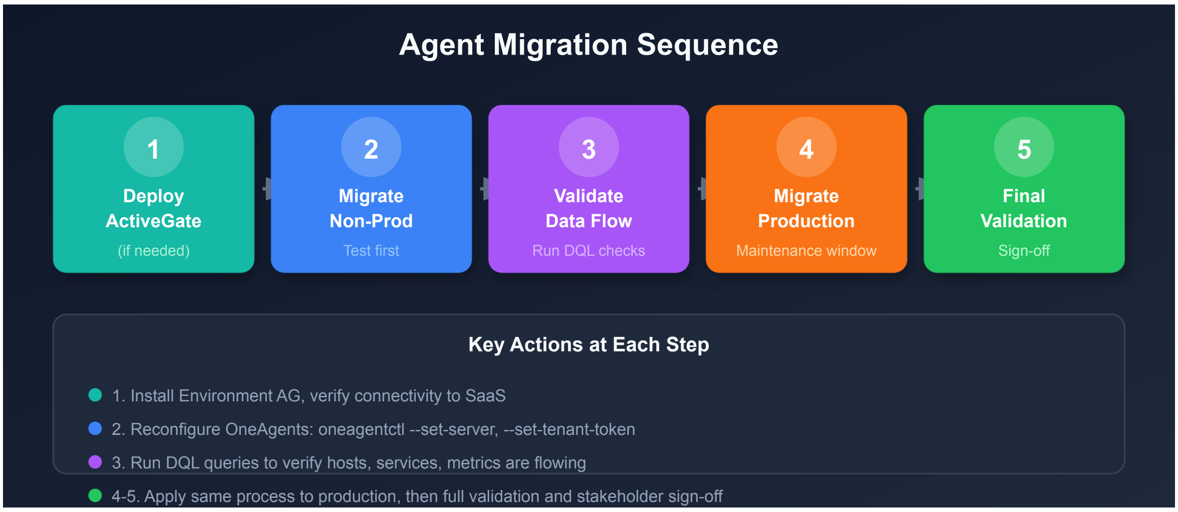
1. Introduction

Migration Options

Option	Method	Best For
Reconfigure	Update existing OneAgent config	Most environments
Reinstall	Fresh OneAgent installation	Major version upgrade
Parallel	Install new, remove old	Zero-downtime requirement

Migration Sequence

1. Deploy Environment ActiveGate (if needed)
2. Migrate non-production hosts first
3. Validate data flow
4. Migrate production hosts
5. Final validation



2. ActiveGate Migration

2.1 When ActiveGate is Required

Scenario	ActiveGate Needed
Hosts can reach SaaS directly	No (optional)
Hosts behind firewall	Yes, for routing
Extensions 2.0 required	Yes
Private synthetic monitoring	Yes
VMware/cloud platform monitoring	Yes

2.2 Deploy Environment ActiveGate

Step 1: Download Installer

From Dynatrace SaaS UI:
1. Go to **Deploy Dynatrace → ActiveGate**
2. Select **Linux** or **Windows**
3. Copy the installer URL and token

Step 2: Linux Installation

```
# Download installer
wget -O Dynatrace-ActiveGate-Linux.sh
"https://{tenant}.live.dynatrace.com/api/v1/deployment/installer/gateway/unix
/latest?Api-Token={token}&amp;arch=x86&amp;flavor=default"

# Run installer
sudo /bin/sh Dynatrace-ActiveGate-Linux.sh
```

Step 3: Windows Installation

```
# Download installer
Invoke-WebRequest -Uri
"https://{tenant}.live.dynatrace.com/api/v1/deployment/installer/gateway/wind
ows/latest?Api-Token={token}" -OutFile Dynatrace-ActiveGate.exe

# Run installer
.\Dynatrace-ActiveGate.exe
```

2.3 Verify ActiveGate Connection

```
// ActiveGate validation - use the Entities API v2 or Settings API
// Note: ActiveGates are not directly queryable via DQL fetch
// Use: GET /api/v2/entities?entitySelector=type("ENVIRONMENT_ACTIVE_GATE")
// Or check the ActiveGates page in the Dynatrace UI
```

---

3. OneAgent Migration Strategies

3.1 Strategy 1: Reconfigure (Recommended)

Update the existing OneAgent to point to SaaS.

Pros:
- No reinstallation required
- Minimal disruption
- Configuration preserved

Cons:
- Requires restart of OneAgent service
- Brief monitoring gap during restart

3.2 Strategy 2: Reinstall

Uninstall Managed OneAgent, install SaaS OneAgent.

Pros:
- Clean installation
- Opportunity to upgrade version

Cons:
- Longer monitoring gap
- More complex process

3.3 Strategy 3: Parallel Installation

Install SaaS OneAgent alongside Managed (temporarily).

Pros:
- Zero monitoring gap
- Easy rollback

Cons:
- Resource overhead during parallel period
- More complex coordination

3.4 Strategy Selection Matrix

Requirement	Recommended Strategy
Minimal downtime	Reconfigure
Major version upgrade	Reinstall
Zero monitoring gap	Parallel
Risk-averse organization	Parallel
Limited change windows	Reconfigure

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4. Migration Execution

4.1 Reconfigure OneAgent (Linux)

Option A: Using oneagentctl

```
# Get current configuration
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --get-server

# Set new SaaS server
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-
server="https://{tenant}.live.dynatrace.com:443/communication"

# Set tenant token (get from SaaS UI)
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-tenant-token="
{tenant-token}"

# Restart OneAgent
sudo systemctl restart oneagent
```

Option B: Using Configuration File

```
# Edit configuration
sudo vi /var/lib/dynatrace/oneagent/agent/config/hostcustomproperties.conf

# Add/update:
# Server=https://{tenant}.live.dynatrace.com:443/communication
# TenantToken={tenant-token}

# Restart
sudo systemctl restart oneagent
```

4.2 Reconfigure OneAgent (Windows)

```
# Get current configuration
&amp; "C:\Program Files\dynatrace\oneagent\agent\tools\oneagentctl.exe" --
get-server

# Set new SaaS server
&amp; "C:\Program Files\dynatrace\oneagent\agent\tools\oneagentctl.exe" --
set-server="https://{tenant}.live.dynatrace.com:443/communication"

# Set tenant token
&amp; "C:\Program Files\dynatrace\oneagent\agent\tools\oneagentctl.exe" --
set-tenant-token="{tenant-token}"

# Restart service
Restart-Service Dynatrace*
```

4.3 Via ActiveGate Routing

If using ActiveGate for routing:

```
# Set ActiveGate as communication endpoint
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-
server="https://{activegate-ip}:9999/communication"

# Set tenant token
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-tenant-token="
{tenant-token}"

# Restart
sudo systemctl restart oneagent
```

4.4 Critical: Application Process Restarts

> 🚨 **IMPORTANT:** After migrating OneAgent, application processes need to be restarted for full monitoring capabilities.

When Restarts Are Required:

OneAgent Mode	Restart Required?	What Happens Without Restart
Full-stack monitoring	✅ **Yes**	Services won't appear; no distributed tracing
Infrastructure-only	✅ **Yes** (for some features)	Application Security vulnerability detection and JMX metrics won't work

Full-Stack Monitoring (Process Restarts Required):

Technology	Why Restart Required
Java	Code modules inject at JVM startup
.NET	Profiler attaches at process start
Node.js	Instrumentation loads at startup
PHP	Zend extension loads at Apache/PHP-FPM start
Go	Instrumentation compiles at build time

What happens without restart:
- Host-level metrics will flow (CPU, memory, disk)
- **Application-level monitoring will NOT work**
- No distributed tracing
- No code-level visibility
- No service detection for existing processes

Infrastructure-Only Mode (Restarts Recommended):

Even in infrastructure-only mode, restarts enable:
- Application Security vulnerability detection
- JMX metrics collection
- Some process-level insights

Recommended Approach:

1. Migrate OneAgent during maintenance window
2. Perform rolling restart of application services
3. Or schedule next regular deployment to restart processes

```
# Example: Restart Java application
sudo systemctl restart myapp.service

# Example: Restart Apache/PHP
sudo systemctl restart apache2

# Example: Restart Node.js application
pm2 restart all
```

4.5 Bulk Migration with Automation

For large environments, use configuration management:

Ansible Example:

```
- name: Reconfigure OneAgent for SaaS
  hosts: all_monitored_hosts
  become: yes
  tasks:
    - name: Set SaaS server
      command: /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-
server="https://{{ saas_tenant }}.live.dynatrace.com:443/communication"

    - name: Set tenant token
      command: /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-tenant-
token="{{ tenant_token }}"

    - name: Restart OneAgent
      systemd:
        name: oneagent
        state: restarted

    - name: Restart application services (for full-stack monitoring)
      systemd:
        name: "{{ item }}"
        state: restarted
      loop: "{{ application_services }}"
      when: restart_apps | default(false)
```

4.6 OneAgent Re-homing Considerations

When using the `oneagentctl` reconfigure approach:

Preserved	Not Preserved
Host ID	N/A (stays same)
Process group IDs	N/A (regenerated based on detection)
Service IDs	N/A (regenerated based on detection)
Custom metadata	✅ Preserved
Environment tags	✅ Preserved
Host group assignment	✅ Preserved
Network zone assignment	Must reconfigure if changed

> **Tip:** Using oneagentctl to reconfigure (rather than reinstall) preserves host identity and custom metadata.

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5. Validation

5.1 Immediate Validation

After migrating hosts, verify they appear in SaaS:

```
// Count hosts reporting to SaaS
fetch dt.entity.host
| summarize totalHosts = count()
```

```
// List hosts reporting to SaaS
fetch dt.entity.host
| fieldsAdd hostName = entity.name
| fields hostName, id
| sort hostName desc
| limit 20
```

5.1.1 OneAgent Version Check

> **Note:** OneAgent version is not available as a DQL field. Use the Dynatrace UI: Navigate to **Manage → Deployment status → OneAgent** to view version distribution.

5.2 Data Flow Validation

Verify metrics are flowing:

```
// Check for recent CPU metrics
timeseries avg(dt.host.cpu.usage), from:-1h, by:{dt.entity.host}
| limit 10
```

```
// Verify logs are flowing
fetch logs, from:-1h
| summarize logCount = count(), by:{time_bucket = bin(timestamp, 5m)}
| sort time_bucket desc
| limit 12
```

5.3 Service Discovery Validation

```
// Check active services discovered after migration
fetch dt.entity.service
| fieldsAdd serviceName = entity.name
| summarize activeServices = count()
```

5.4 Migration Completion Checklist

Validation	Query/Method	Status
All hosts reporting	Entity count matches expected	[ ]
CPU metrics flowing	Timeseries query returns data	[ ]
Memory metrics flowing	Timeseries query returns data	[ ]
Logs ingesting	Log count > 0	[ ]
Services discovered	Service count matches expected	[ ]
No OneAgent errors	Check OneAgent logs	[ ]

Troubleshooting

Common Issues

Issue	Symptom	Solution
Connection refused	Host offline in SaaS	Check firewall rules
Invalid token	OneAgent errors in log	Regenerate tenant token
DNS resolution	Can't reach endpoint	Verify DNS settings
Certificate errors	SSL/TLS failures	Check proxy configuration
ActiveGate not found	Routing failures	Verify AG is running

OneAgent Log Locations

Platform	Log Path
Linux	`/var/log/dynatrace/oneagent/`
Windows	`C:\ProgramData\dynatrace\oneagent\log\`

Rollback Procedure

If migration fails, revert to Managed:

```
# Revert to Managed server
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-
server="https://{managed-cluster}/communication"
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-tenant-token="
{managed-token}"
sudo systemctl restart oneagent
```

---

6. Next Steps

Immediate Actions

1. **Deploy ActiveGate** - If required for your architecture
2. **Migrate test hosts** - Start with non-production
3. **Validate thoroughly** - Run all validation queries
4. **Migrate production** - After successful test
5. **Document issues** - Track any problems encountered

Continue the Series

Next Notebook	Focus
M2S-07: Security & Privacy	Security considerations for SaaS

Agent Resources

- OneAgent Documentation
- ActiveGate Documentation
- OneAgentCtl Reference

Summary

In this notebook, you learned:

- How to deploy Environment ActiveGates for SaaS
- OneAgent migration strategies and when to use each
- Step-by-step reconfiguration procedures
- Validation queries to confirm successful migration
- Troubleshooting common migration issues

> **Key Takeaway:** The reconfigure strategy is simplest for most migrations. Use oneagentctl to update the server endpoint and tenant token, then restart the service.

Continue to M2S-07: Security & Privacy for security considerations in SaaS.
